

MAHFOUDI SAAD

Embedded Systems Engineer

saad.ma7f@gmail.com | +212 623 659 700 | Casablanca, Morocco | [linkedin.com/in/saad-mahfoudi](https://www.linkedin.com/in/saad-mahfoudi) | Driving licence: Category B

High-voltage test and validation engineer specialised in electric and hybrid propulsion systems. My experience spans HV validation at Stellantis (through Capgemini Engineering), R&D test-tool development for automotive embedded systems, and projects in power electronics, embedded systems, and ADAS. I am proficient in MATLAB/Simulink, LTspice, PSIM, embedded programming in C/Python, and the CAN, LIN, and UART protocols. My goal is to contribute to the development of reliable and safe electrified mobility solutions.

EDUCATION

Engineering Degree — Electrical Engineering & Industrial Control

Hassan II University, Faculty of Sciences and Techniques (FST) Mohammeda | 09/2021 – 07/2024

Professional Bachelor's Degree — Diagnostics & Maintenance of Automotive Embedded Electronic Systems

Mohammed V University, École Supérieure de Technologie (EST) Salé | 09/2020 – 07/2021

University Diploma of Technology (DUT) — Electrical Engineering & Industrial Computing

Mohammed V University, École Supérieure de Technologie (EST) Salé | 10/2018 – 07/2020

PROFESSIONAL EXPERIENCE

High Voltage Safety Test & Validation Engineer

Capgemini Engineering (Client: Stellantis) | 08/2025 – Present | Casablanca

As a member of the High Voltage Safety Integration team at Capgemini Engineering for client Stellantis, I am responsible for testing and validating the quality of propulsion networks in hybrid and electric vehicles. My work focuses on ensuring the safety, reliability, and compliance of high-voltage systems with industry standards. I actively contribute to the development of efficient and safe electrified mobility solutions by conducting in-depth analyses of high-voltage component performance, validating safety protocols, and optimising the integration of electric propulsion systems according to regulatory requirements and Stellantis client specifications.

Key Skills: Propulsion systems validation; Electric/hybrid vehicle safety; Industrial standards compliance; Time- and frequency-domain analysis of battery & high-voltage network performance; Signal processing; LTspice; MATLAB/Simulink/Simscape

Internal R&D Project — BMS Proof of Concept (PoC)

Capgemini Engineering | Internal project

Contributed to the development of a Battery Management System (BMS) Proof of Concept applied to a 48V/100Ah LiFePO4 battery pack. The project covered the full development cycle, from requirements to prototype validation:

- Took part in defining and implementing the PoC's technical requirements and functional specifications
- Modelled the battery pack in MATLAB/Simscape: cell selection (LiFePO4 chemistry), series/parallel configuration sizing for the 48V/100Ah target, and electrochemical behaviour simulation
- Designed and modelled the BMS architecture: cell balancing, SOC/SOH estimation, and protection logic
- Selected hardware components for the physical prototype design
- Developed and validated embedded firmware on an STM32 Nucleo board (C programming, real-time management, communication interfaces)
- Carried out integration testing and validation of the complete BMS prototype

Key Skills: BMS design; LiFePO4 battery modelling; MATLAB/Simscape; STM32 Nucleo; Embedded C firmware; Requirements engineering; Prototype testing & validation

Embedded Electronic Systems Engineer — R&D Test Tools

NGE Automotive | 11/2024 – 08/2025 | Casablanca

Responsible for researching, designing, and developing test tools for automotive embedded systems. Designed electronic circuits and PCBs. Selected and integrated microcontrollers and components suited to automotive applications. Developed embedded software in C/C++, covering microcontroller programming, driver and middleware development, and real-time system

management. Proficient in communication protocols (UART, SPI, LIN, CAN/CAN FD) for system integration and diagnostics. Implemented advanced techniques such as PIO programming (RP2040) and DC/DC converter design.

Key Skills: PCB; UART, LIN, CAN, USB; Firmware development in C; Hardware and software debugging; Oscilloscopes; Low-level programming and interrupt management; DC/DC converter design and optimisation; Advanced techniques (e.g. PIO programming on RP2040); PCAN-View; BUSMASTER; Muxtrace

Graduate Internship (End-of-Studies Project)

Capgemini Engineering | 02/2024 – 07/2024 | Casablanca

Project objective: develop a simulation and data-visualisation tool for an electric vehicle battery test bench for Stellantis, to verify and analyse battery pack performance against client specifications. The system:

- Simulates real-condition test data in MATLAB/Simulink
- Visualises results (voltage, current, SOC, SOH, temperature, etc.) through an interactive dashboard
- Validates battery compliance with client technical requirements
- Optimises testing by reducing development cost and time

Key Skills: MATLAB/Simulink; Stateflow; Arduino; Proteus/ARES; Lithium-ion battery; BMS

Final-Year Internship

FST Mohammedia | 05/2023 – 07/2023 | Mohammedia

Project objective: design and build a vehicle prototype with ADAS functionality. Developed ADAS features (traffic-sign and traffic-light detection, lane tracking). Integrated detection modules into the embedded system; carried out unit testing and validation.

Key Skills: OpenCV; Raspberry Pi; ADAS; Python; Artificial Intelligence; IP addressing; Proteus

SKILLS

Power electronics: DC/DC; AC/DC; DC/AC; AC/AC

Signal acquisition & processing: Laplace transform; Fourier transform/series; filtering; level shifting

Electronic design: PCB design: KiCad; Altium; Proteus; Eagle; PSIM; LTspice

Embedded systems: PIC16F84, 877; STM32; Raspberry Pi (models 3, 4, 5 — ARM CPU); Raspberry Pi Pico (RP2040, ARM Cortex-M0+); RISC-V; Radxa Rock 3C

Communication protocols: CAN; CAN FD; LIN; UART; SPI

Simulation & development software: PSIM; MATLAB/Simulink/Simscape/Stateflow; LTspice; Proteus

Programming languages: C/C++; MikroC; Assembly; Python; VHDL

Networking & infrastructure: Cisco; network configuration and administration

Office tools: Microsoft Office Suite: Excel; Word; PowerPoint

STRENGTHS

- Rigour and autonomy
- Adaptability and proactivity
- Drive and team spirit

INTERESTS

- Volunteering: Robothing club
- Audiovisual: video editing / design
- Self-development: personal growth

LANGUAGES

French: Fluent

English: Fluent

Arabic: Native